



An ant-sized micro-bot that travels in swarms (Credit: popsci.com)

## How micro-bots function

When layers of platinum and titanium are put on a silicon wafer and electrical voltage is applied, platinum contracts while titanium remains rigid, bending the flat surface, which becomes the motor that moves the limbs of the robots. Sensors are used to measure temperature or electrical pulses. Micro-bots use the same basic technology as computer chips. A hundred-micron silicon chip can contain millions of micro-bots with robot limbs.

Micro-bots are usually powered by shining lasers on tiny solar panels fixed on their backs and run on a fraction of a volt consuming ten-billionths of a watt. However, for micro-bots injected into the brain or body, system lasers would not work as the power source, so magnetic fields are used as an alternative. Micro-bots can usefully be integrated into small systems, such as micro-electromechanical systems (MEMSes) and nano-electromechanical systems (NEMSes), to provide the advantage of reduction in required resources (mass, volume, power, etc.) and the overall cost of the systems.

Micro-bots working together in large numbers create a dense network of parallel, self-organising, reconfigurable swarm. These enable applications in the field of human healthcare, such as minimally-invasive diagnosis and treatment inside the human body; biological studies or bio-

engineering applications inside microfluidic devices; desktop micro-manufacturing; and mobile sensor networks for environmental and health monitoring. Micro-bots perform targeted therapy by navigating in the cardiovascular system, and have the potential for providing minimally-invasive surgeries and treatments through early diagnosis.

## Micro-bots at work in swarms.

It is possible to imitate insects and worms and develop actuators that effectively reproduce the behaviour of their natural muscle, and have excellent mobility and dexterity. Dielectric elastomers are used, which have the ability to replicate many features of natural muscle.

Rolled actuators are made in a variety of sizes and shapes similar to those of a natural muscle with both silicone and acrylic materials. These have high strains and are flexible to enable motion over rough surfaces.

Robotic platforms using an inch-worm-like propulsion system could be used for tasks such as inspection in narrow pipes.

These micro-bots are mostly made of rubber and are resistant to shock and abuse, in sharp contrast to brittle and delicate micro-machined silicon-based devices. Although, individually, micro-bots are small and weak, a number of these acting together in a swarm can cover an area faster and more reliably. These can be transport-

ed in space probes to fan out and study the surfaces of planets or moons. Such micro-bots allow extending human approach to locations that are dangerous, dirty or difficult to reach.

## Application areas of micro-bots

Micro-bots have a wealth of interesting applications such as autonomously operating inside the human body using natural pathways like blood vessels or gastrointestinal tract for diagnosis and treatment, and delivering drugs to a particular area of the body (called targeted medicine); burrowing into the brain to measure nerve signals; cleaning and rejuvenating cellphone batteries; monitoring and reporting in different environments; testing ideas in networking and communications; and deployment as drones.

Micro-bots are increasingly being used by engineers and scientists leading to further innovations in the fields of engineering, biology and medicine, surgery, defence, environment, etc. Generally, these applications include operation at small scales with the help of high-precision micro-bots in such areas as nanoscale assembly and manufacturing; manipulation of nanoscale objects; mechanical and electrical characterisation at nanoscale; structuring of nano-materials; measurements requiring positioning with nano-scale precision; and many others.

**Micro-robotics in biology and medicine.** Synthetic biology programmable sensors with memory and signal processing units are used to build genetically-engineered bacteria

with magnetic micro-robots. Micro-robots called artificial bacterial flagella (ABF) resemble the natural bacteria Flagella and have a magnetic head.

These can be directed by varying the strength and direction of the rotating magnetic field for targeted delivery deep inside the human body. Micro-bots are useful in drug delivery and fighting diseases such as cancer and heart diseases with

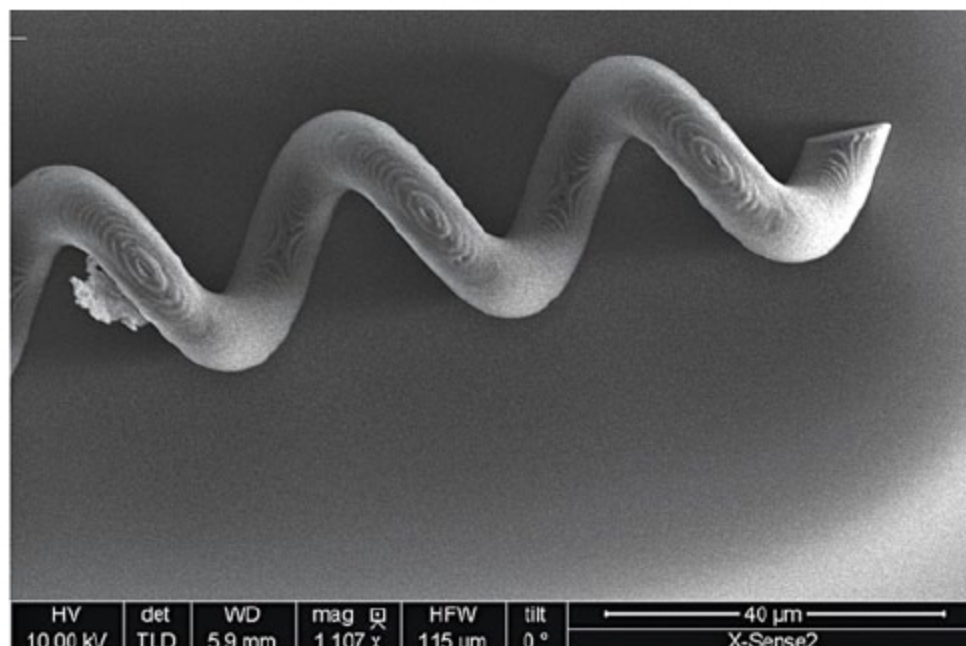


Image of an artificial bacterial flagella (ABF) (Credit: researchgate.net)